CAN Signal Matrix - Display CAN, Pioneer Standard CAN and V2

Status: Working evidence matrix

Last updated: 2026-08-06

Audience: Firmware developer, decoder reviewer and hardware tester

Purpose and evidence levels

This document separates signals by physical bus and vehicle/profile source. It must not be read as proof that similarly named values are interchangeable.

Status	Meaning
Confirmed by MOT	Reproduced during a controlled live capture on the Microlino Pioneer
Plausible	Raw values and behaviour fit the proposed interpretation, but meaning or scale still needs an independent check
Externally confirmed for V2	Supplied by a separate V2 reverse-engineering source; MOT has no V2 vehicle for direct validation
Externally indicated	Present in supplied notes/screenshots, but byte order, scale, signedness or exact semantics remain incomplete
Not observed	Targeted ID did not appear during the stationary Pioneer capture

All MOT captures used two ESP32-C6 TWAI controllers at 500 kbit/s in listen-only mode. CAN1 was connected to Standard CAN and CAN2 to Display CAN.

Cross-bus signal summary

Signal	Display CAN	Pioneer Standard CAN (V1 vehicle)	Standard CAN V2 evidence	Current assessment
Display SOC	`0x602 data[0] / 2`	No equivalent confirmed	`0x18D data[7]` according to external source	Display CAN confirmed by existing decoder. Pioneer `0x18D data[7]` stayed 90 while the display moved from 97 to 100%, so it is not Pioneer Display SOC.
Standard/BMS SOC candidate	Not used	`0x18D data[7]`	`0x18D data[7]`	Raw Pioneer field confirmed, semantics unresolved; externally confirmed as Display SOC for V2. Keep sources separate.
Speed	`0x602 data[1] / 2`	Not found	No supplied mapping	Display-CAN decoder field.
Odometer	`0x602 data[4..7]` with current packed scaling	Not found	No supplied mapping	Display-CAN decoder field.
HV pack voltage	Not currently decoded	`0x18D data[3..4]`, unsigned little-endian mV	Same mapping	Confirmed by MOT on Pioneer and externally confirmed for V2.
Battery current	Not currently decoded	`0x18D data[1..2]`, signed little-endian; `raw x 0.3 A`	External material also identifies a current field and lists `0x2BA`	Scale and sign confirmed on Pioneer during multi-level charging, traction and regeneration.
DC charging power	Dashboard power byte is unscaled and bidirectional	Derived as pack voltage x battery current	Derivable if current scaling is shared	Pioneer result agrees closely with two measured AC-input points after allowing for losses.
Plugged	Existing candidate `0x604 data[4] & 0x10`	`0x18D data[6]`: stable `0x10` unplugged and `0x20` plugged	`data[6]` externally described as charge/status	Pioneer Standard-CAN state confirmed, including an AC-deenergized EVSE test. Display-CAN candidate still needs a direct state comparison.
Active charging	Current `0x603` power-threshold inference is not reliable	Positive signed `0x18D data[1..2]`; idle offset about -3/-4	Current signal externally indicated	Confirmed on Pioneer across start, stable charge, AC removal and natural completion.
Cell values A/B	Not decoded	`0x4AD data[0..1]` and `[2..3]`, unsigned little-endian mV	Same mapping	Plausible on Pioneer in stable states; externally confirmed for V2. Exact semantics as individual cells versus reported extrema remain open.
Cell min/max/delta	Not decoded	Derived from the two `0x4AD` values	Same derivation	Plausible on Pioneer; externally confirmed for V2.
SOH	Not decoded	Not found	`0x1B1 data[2..3] / 100` candidate	Externally indicated only.
Pedal/power request	`0x603 data[4]` is the dashboard power display	`0x37F data[2..3]` candidate	Same candidate	`0x37F` was present but constant in stationary Pioneer testing; driving evidence is required.

Pioneer Standard-CAN findings

CAN ID `0x18D`

Bytes	Type and conversion	Proposed meaning	Pioneer evidence	Same as supplied V2 evidence?
`data[0]`	`uint8`	rolling counter or sequence	Covered nearly the full byte range during stable charging; not a direct measurement	Unknown
`data[1..2]`	`int16`, little-endian	battery current; amperes = raw x 0.3	Scale and sign confirmed during charging, traction and regeneration	Likely related, but the V2 source does not expose an unambiguous raw-ID association
`data[3..4]`	`uint16`, little-endian mV	HV pack voltage	Confirmed by MOT	Yes
`data[5]`	`uint8`	unknown	Stable `0x00` in recorded phases	Unknown
`data[6]`	`uint8` state	`0x10` unplugged; `0x20` plugged	Confirmed by repeated powered and AC-deenergized plug tests; transient `0x04` also occurred	V2 source calls it a charge/status byte
`data[7]`	`uint8`	Standard/BMS SOC candidate	Raw value confirmed; stayed 90 while Pioneer display showed 97-100%	Same byte, different observed Pioneer semantics

Charging evidence for `0x18D data[1..2]`

Vehicle/charger state	Signed raw	Pack voltage	External measurement	Interpretation
Awake, unplugged	about -4	about 53.0 V	0 W	Idle offset
Plugged, EVSE AC-side off	about -4	about 53.0 V	0 W	Plugged but not charging
Stable charge	112-118	about 53.3 V	1,836 W gross AC input	`115 x 0.3 A x 53.3 V ≈ 1,839 W`
Taper charge at 98-99% display SOC	43-47	about 53.2 V	762 W gross AC input	`45 x 0.3 A x 53.2 V ≈ 718 W` before conversion losses
External AC removed, cable retained	about -4	falling from charge voltage	0 W	Charging stopped, still plugged
Natural completion at 100% display SOC	fell from +45 to -3	about 53.24 V at end	Fell from 762 W to 0 W	Stable post-charge idle offset

Lower-SOC measurements at several charge settings confirmed 0.3 A per raw unit. At about 50 V, the derived DC values tracked independent 1,809 W, 2,088 W and approximately 2,340 W references within expected conversion and vehicle-load differences. Three controlled road captures then confirmed negative current during traction and positive current during regeneration. Observed results included about 4.8 kW pedal-release regeneration, 7.8 kW with light braking and a later normal- road peak of about 20.9 kW discharge.

CAN ID 0x4AD

Bytes	Type and conversion	Proposed meaning	Status
`data[0..1]`	`uint16`, little-endian mV	cell value A	Plausible on Pioneer; externally confirmed for V2
`data[2..3]`	`uint16`, little-endian mV	cell value B	Plausible on Pioneer; externally confirmed for V2
derived	`min(A, B)`	minimum cell value	Plausible
derived	`max(A, B)`	maximum cell value	Plausible
derived	`max(A, B) - min(A, B)`	cell delta	Plausible

Stable examples included 4,081/4,078 mV before charging and approximately 4,105/4,100 mV during charge. Larger transient divergence occurred around charge start, so the exact semantics must remain provisional until repeated captures exclude multiplexing or transition-specific payload use.

Other targeted Pioneer IDs

CAN ID	Pioneer result	External V2 information
`0x101`	Not observed	Byte 4 candidate for positive power request; unit/scale incomplete
`0x107`	Not observed	Separate ESPHome source watched bytes 0-3 without a decoding rule
`0x1B0`	Not observed	Bytes 2-3 pack voltage, 4-5 minimum cell voltage, 6-7 maximum cell voltage; byte order incomplete
`0x1B1`	Not observed	Bytes 0-1 SOC x100, bytes 2-3 SOH x100; byte order incomplete
`0x2BA`	Not observed	Bytes 0-1 battery-current candidate; signedness/order/exact scale incomplete
`0x37F`	Observed and constant while stationary	Bytes 2-3 pedal/power-request candidate

Display-CAN signal map

CAN ID 0x602

Bytes	Current decoding	Status
`data[0]`	SOC = byte / 2	Implemented Display-CAN signal
`data[1]`	speed = byte / 2 km/h	Implemented Display-CAN signal
`data[4..7]`	packed odometer / 1024 km	Implemented Display-CAN signal
derived	estimated range from SOC	Firmware-derived estimate, not a direct vehicle signal

CAN ID `0x603`

The external source uses one-based byte and bit numbering. Positions below are converted to zero-based firmware indices, assuming its bit 1 is the least- significant bit.

Firmware position	Mask	Meaning	Status
`data[1]`	`0x01`	seat-belt sensor/status	Externally indicated
`data[2]`	`0x08`	handbrake	Externally indicated
`data[2]`	`0x02`	parking/position light	Externally indicated
`data[2]`	`0x04`	low beam	Externally indicated
`data[3]`	`0x80`	D/R interlock warning when brake was not pressed first	Externally indicated
`data[3]`	`0x20`, `0x40`	indicator-state bits	Externally indicated; left/right mapping missing
`data[4]`	full byte	dashboard power display	Implemented; sign/scale and charging interpretation remain provisional
`data[5]`	`0x01`	sport mode	Externally indicated
`data[6]`	`0x02`	brake pressed	Externally indicated

`data[4]` is a bidirectional dashboard display value. It must not by itself be used as proof of active charging. The current firmware threshold-derived charging state is compatibility behaviour pending replacement by the confirmed Standard- CAN current/plug signals or another independently verified source.

CAN ID `0x604`

Firmware position	Mask	Meaning	Status
`data[4]`	`0x10`	plugged candidate	Implemented, but not yet directly compared with the controlled plug sequence

Decoder policy

- Preserve Display-CAN and Standard-CAN values as source-specific telemetry.
- Do not overwrite Display-CAN SOC with `0x18D data[7]` on Pioneer.
- Permit every registered decoder profile on every physical CAN controller; board defaults are configuration policy, not decoder-engine restrictions.
- Keep CAN operation listen-only during discovery and pilot operation.
- Promote plausible or externally indicated fields to confirmed only after a repeatable state change or independent physical reference.

Remaining validation

- Stationary/drive comparison of `0x37F` to test the pedal hypothesis.
- Direct Display-CAN `0x604` plug-state comparison.
- V2 vehicle capture when access becomes available.
- Controlled cross-channel injection and extended dual-CAN soak.

Related documents

- CAN and decoder pipeline
- Firmware architecture
- C6-001 dual-CAN sprint